

Hands-on Activity 3.1	
Creating an Interactive Face Filter	
Course Code: CPE 411	Program: Computer Engineering
Course Title: Interactive Extended Reality	Date Performed: 05/03/2026
Section: 22S2	Date Submitted: 05/03/2026
Name(s): Bulaong, Ross Andrew D.	Instructor: Engr. Jimlord M. Quejado
Esguerra, Serge	
Madriago, Dave	
Palomares, Charles Yexel	
Reyes, Jamie Rose	

1. Discussion
This activity guides learners in making an Interactive AR Face Filter similar to those in Tiktok and Instagram
2. Materials and Equipment
Unity Hub and Unity Editor Learn Unity Website (Unity Essentials Pathway - Learn Game Development for Beginners Unity Learn) Computer with Network Connection
3. Output

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BULAONG, ROSS ANDREW

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Accessibility Settings

Use High Contrast UI

Use a Dyslexia Friendly Font

Extended Reality > Assignments

Interactive Face Filter

Immersive Reader

Activity 3.1 Creating an e Face Filter

12:30p.m. Points 24 Submitting a file upload at 12a.m. - Feb 26 at 1:30p.m.

you've learned with filters that add interactive elements. You'll be customizing and trying on glasses, then an ocean-themed face filter.

the given mission in the unity pathway: Creating an Interactive Face Filter. You'll demonstrate its use. Demonstrate its use in a 10-second video. Upload the short video and screenshots in a lab report.

Lifelong Learning

Ratings			
6 pts Excellent Educational interests and pursuits exist and flourish outside classroom requirements, knowledge and/or experiences are pursued independently and applies knowledge	5 pts Good Educational interests and pursuits exist and flourish outside classroom requirements, knowledge and/or experiences are pursued independently	4 pts Satisfactory Look beyond classroom requirements, showing interest in pursuing knowledge	3 pts Unsatisfactory Beginner level, lack of interest in pursuing knowledge

Basic Face Filters - InteractiveFaceFilter - Android - Unity 2022.3.62EP1 <DK11>

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Hierarchy

Scene

Game

Project Settings

Inspector

Interac: 1

XR Orig

AR Sess

Direction

Glass 3

ARJC 3

VisualSc

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Project

Console

Assets

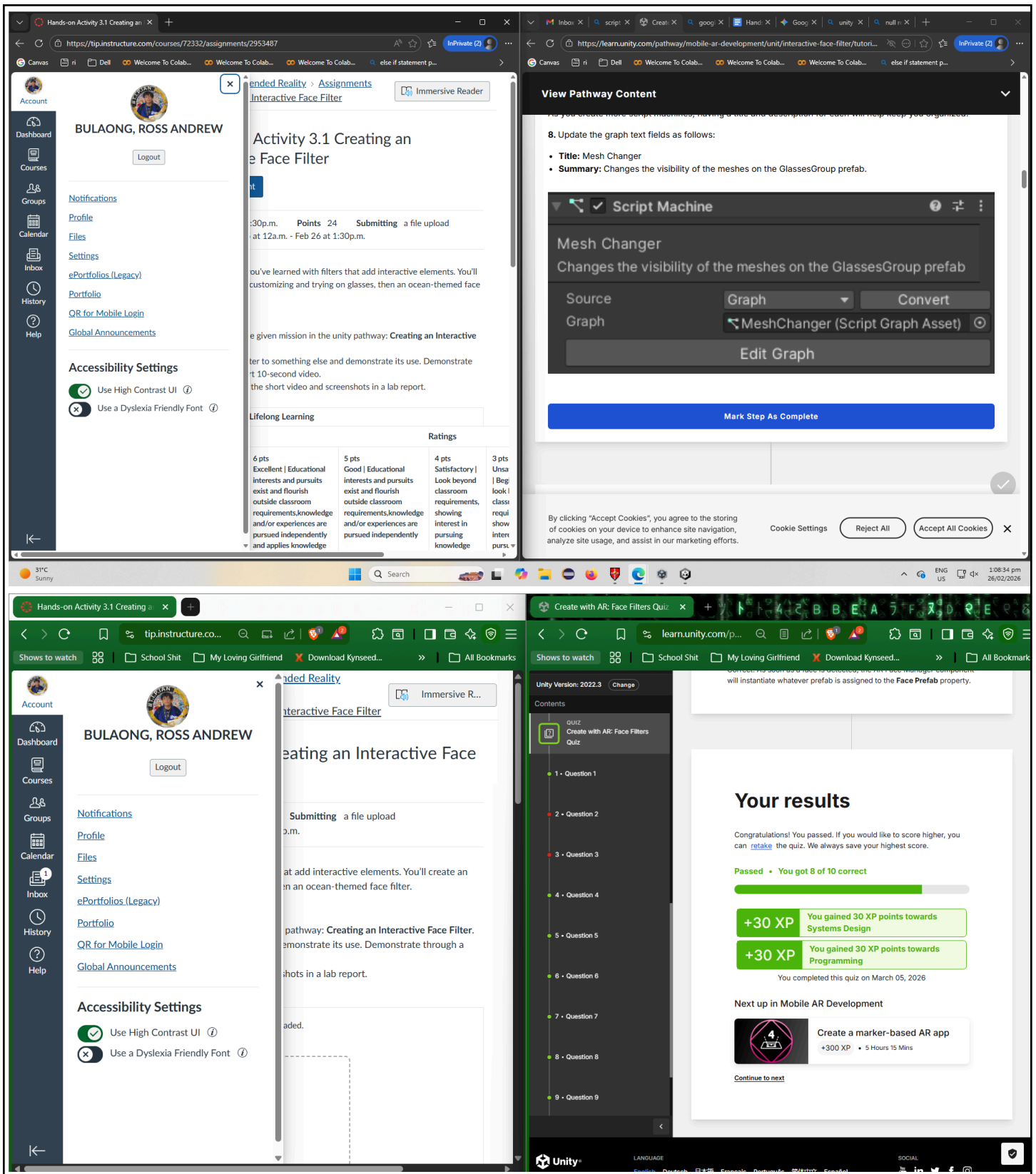
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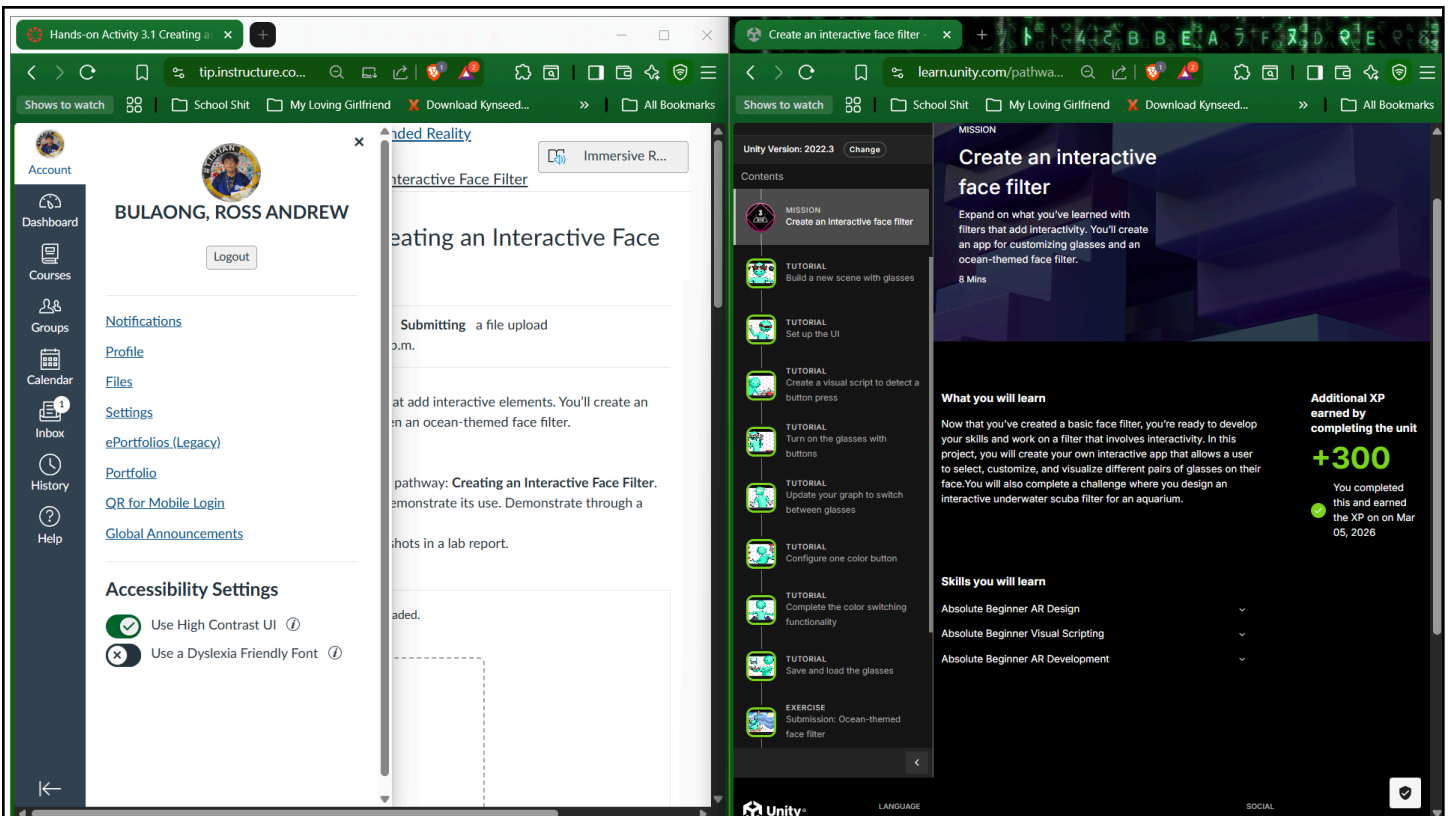
Project

Console

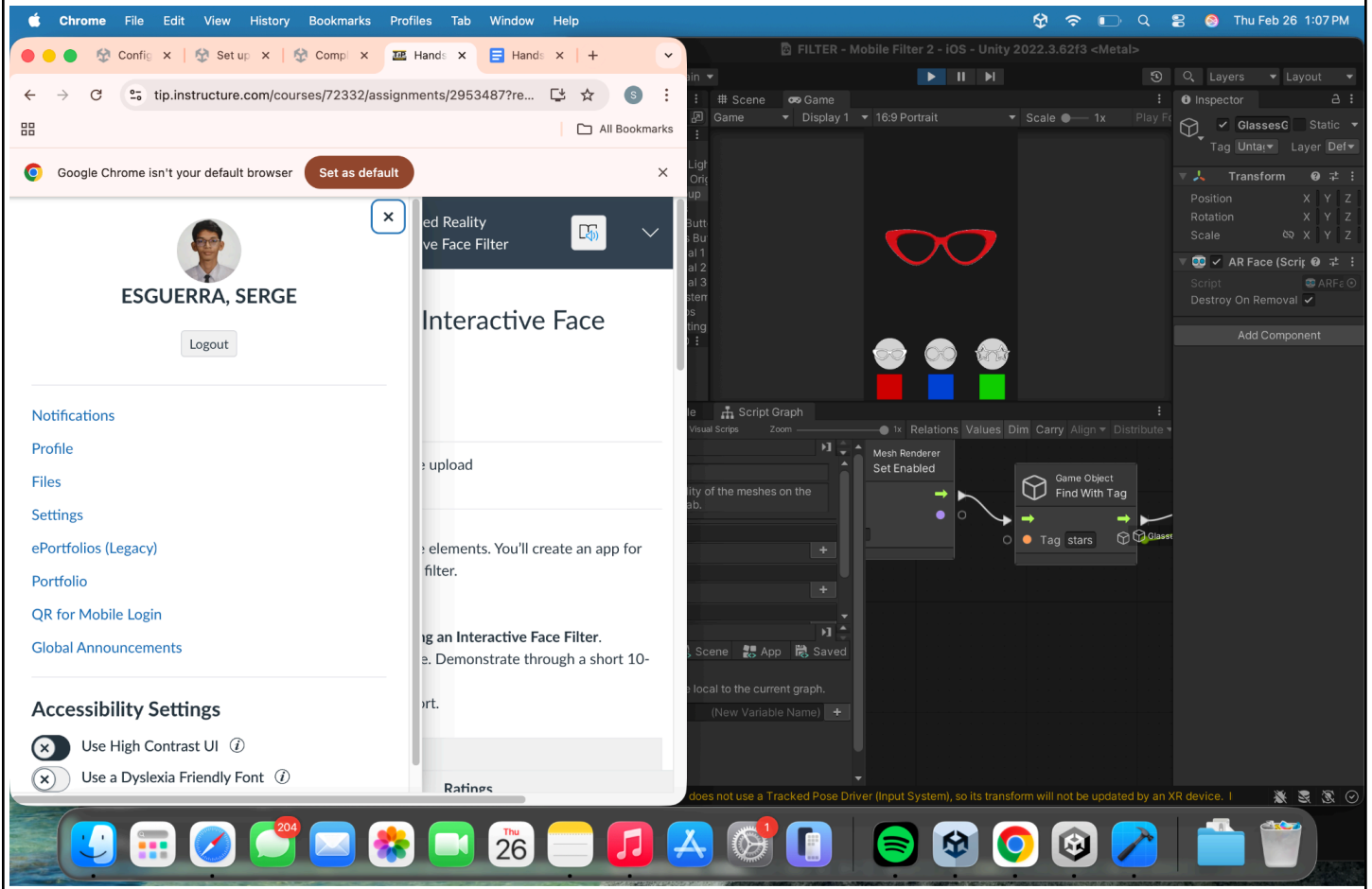
Assets

Assets





ESGUERRA



User Profile



Edit Profile Picture

ESGUERRA, SERGE

Contact

No registered services, you can add some on the [settings](#) page.

Biography

No biography has been added

Links

No links have been added

View Pathway Content

MISSION

Create an interactive face filter

Expand on what you've learned with filters that add interactivity. You'll create an app for customizing glasses and an ocean-themed face filter.

8 Mins

Additional XP earned by completing the unit

+300

✓ You completed this and earned the XP on on Mar 05, 2026

✓

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XR Origin AR R x User Profile x +

tip.instructure.co... ☆

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Hands-on Activity 3.2 - Creat x Create an interactive face filt: x JohnDonald messaged Regis: x +

learn.unity.com/pathway/mobile-ar-development/unit/interactive-face-filter?version=2... ☆

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Your results

Congratulations! You passed. If you would like to score higher, you can [retake](#) the quiz. We always save your highest score.

Passed • You got 10 of 10 correct

+30 XP You gained 30 XP points towards Systems Design

+30 XP You gained 30 XP points towards Programming

You completed this quiz on March 05, 2026

Next up in Mobile AR Development

4

Create a marker-based AR app

+300 XP • 5 Hours 15 Mins

✓

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Hands-on Activity 3.1 Creating

Create with AR: Face Filters Quiz

learn.unity.com/pathway/mobile-ar-development/unit/interactive-face-filter/quiz/create-with-ar-face-filters-quiz

Correct. As soon as a face is detected, the AR Face Manager component will instantiate whatever prefab is assigned to the **Face Prefab** property.

Contents

QUIZ

Create with AR: Face Filters Quiz

1 • Question 1

2 • Question 2

3 • Question 3

4 • Question 4

5 • Question 5

6 • Question 6

7 • Question 7

8 • Question 8

9 • Question 9

10 • Question 10

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Next up in Mobile AR Development

4

Create a marker-based AR app

+300 XP • 5 Hours 15 Mins

[Continue to next](#)

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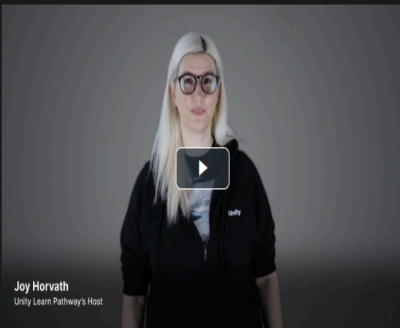
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devices:

Continue Pathway → Last visited on March 05, 2026

Learners that completed pathway: 1509



Joy Horvath
Unity Learn Pathways's Host

Level: Foundational

Pathway completion bonus: +600 XP

Completion time: 8 weeks

Pathway Description

Welcome to AR Development! In this learning Pathway, you'll create Augmented Reality (AR) applications that are compatible with iOS and Android devices. You'll use Visual Scripting and Unity's AR tools to develop a wide variety of engaging AR experiences. This pathway assumes a basic knowledge of Unity.

Skills you will learn

Unity Version 2022.3

56% complete

MISSION 1 Get started with AR

MISSION 2 Create a basic face filter

MISSION 3 • COMPLETED MARCH 05, 2026 Create an interactive face filter

8 hours 35 mins

Expand on what you've learned with filters that add interactivity. You'll create an app for customizing glasses and an ocean-themed face filter.

Mission Details

Tutorial • Build a new scene with glasses +10 XP

Tutorial • Set up the UI +10 XP

Tutorial • Create a visual script to detect a button press +10 XP

Tutorial • Turn on the glasses with buttons +10 XP

Tutorial • Update your graph to switch between glasses +10 XP

Tutorial • Configure one color button +10 XP

Tutorial • Complete the color switching functionality +10 XP

Tutorial • Save and load the glasses +10 XP

Exercise • Submission: Ocean-themed face filter +60 XP

Quiz • Create with AR: Face Filters Quiz +60 XP

+300 XP Additional XP for completing the mission

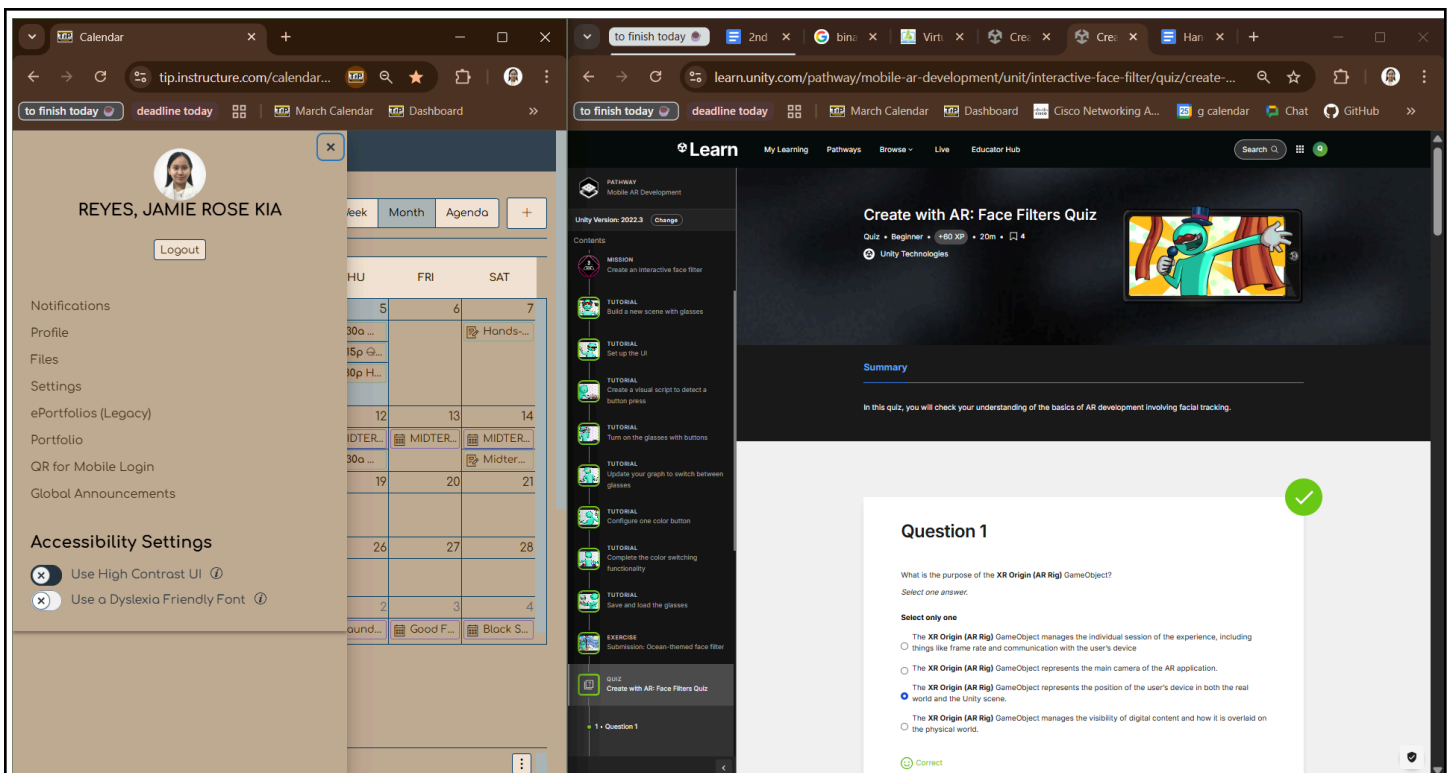
COMPLETION SCREENSHOT: MARCH 5, 2026

REYES

The image is a composite screenshot showing three main components:

- Top Left:** A web browser window displaying a user profile for "REYES, JAMIE ROSE KIA" with a "Logout" button. The browser address bar shows "tip.instructure.com".
- Top Right:** A web browser window displaying a tutorial page titled "Update your graph to switch between glasses" from the Unity Learn platform. The page includes a video player showing a person wearing glasses and a list of steps for the tutorial.
- Bottom:** A screenshot of the Unity 2022.3.62f3 interface. The Hierarchy panel shows the "InteractiveFilter" object with various sub-objects like "XR Origin (AR Rig)", "AR Session", "Directional Light", "GlassesGroup", "Glasses_Cat", "Glasses_Simple", "Glasses_Stars", "AR_Canvas", "Glasses Buttons", "Materials Buttons", "EventSystem", "Visual Scripts", "VisualScripting SceneVariables", "XRInteractionManager", and "DontDestroyOnLoad". The Inspector panel shows the "Glasses" component with various settings. The Console panel shows a message: "No active UnityEngine.XR.ARSubsystems.XRRaycastSubsystem is available. This feature is either not supported on the current platform, or you may need to enable a provider in Project Settings > XR Plug-in Management."

FINAL SCREENSHOT: MARCH 5, 2026



Conclusion

Bulaong, Ross Andrew D.:

Completing this activity was incredibly challenging and a brain-buster activity for me. When I was in the school's laboratory, the Visual Scripting tool kept failing on me even with resets, updated packages, and different AI agents helping me fix the unfixable error! It was a headache, but even in those moments, I learned an incredibly important skill both as a game developer and a computer engineer: patience. As I got home, I retried and finally completed the Unity Learn pathway without any errors. The ocean-themed AR filter provided by Unity Learn as the final exercise made me squeeze out the skills I've learned prior to the activity, which improved my technical and scripting skills.

Reyes, Jamie Rose Kia M.:

Completing this activity was both challenging and rewarding for me. At first, learning how to build interactive face filters felt overwhelming, but as I went through each step, from setting up the scene and adding elements to designing UI and using visual scripting, I gradually became more comfortable with the process. Along the way, I encountered an issue in visual scripting where the feature did not work as expected because of a small typo in setting an argument, which taught me the importance of paying attention to details and carefully checking my work. The final ocean-themed filter activity helped me reflect on how much I had learned, as it brought all the concepts together into one project. Overall, this pathway not only improved my technical skills but also gave me more confidence in creating creative AR experiences.

Palomares, Charles Yexel:

I learned a lot in doing this activity, like how to make things move with visual scripting and how to track faces in AR. I make interactive face filters with many glasses and materials, and also ocean theme filters. I also had many problems though, buttons broke, face tracking failed, and the filter didn't show up. But after much trial and error, I finally made it work! This activity helped me learn much more about AR development. Thus, I will use what I have learned to create an interactive face AR filter, let others use it so they can have fun and also gain experience for my future career.

Esguerra, Serge

Doing this activity was a fun and challenging learning experience for me, as I was able to create an interactive face filter that displays different kinds of glasses that can be customized with different materials, and an Ocean_themed face filter. I was able to learn how to create filters that can interact with the user, how to use visual scripting, and apply previous AR development experiences where objects or faces can be tracked in the real world. I was also able to encounter many errors, such as the face tracking and buttons not

